

Andrei-Cătălin Popa

EMJM ASTROPHYSICS AND SPACE SCIENCE STUDENT

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Education

Université Côte d'Azur

Nice, France

MSC IN ASTROPHYSICS AND SPACE SCIENCE (ERASMUS MUNDUS JOINT MASTER'S PROGRAMME)

Mar. 2025 – present

- Jointly delivered by a consortium of four universities: Università degli Studi di Roma "Tor Vergata", Universität Bremen, Université Côte d'Azur, and University of Belgrade.
- Thesis:** *Characterising M-dwarf Stars as Exoplanet Hosts in preparation for the PLATO mission.*
Advisors: Dr. Orlagh Creevey (Observatoire de la Côte d'Azur), Dr. Valentina D'Orazi (Università di Roma Tor Vergata) and Dr. Ulrike Heiter (Uppsala University).
- Internship:** *Fundamental Parameters of FGK stars.*
Advisors: Dr. Orlagh Creevey (Observatoire de la Côte d'Azur) and Dr. Caroline Soubiran (Laboratoire d'Astrophysique de Bordeaux).
- Key modules:** Astronomical Techniques, Astronomical Optics, Helio- & Asteroseismology, Space Plasmas, Exoplanets, Space Missions & Design, Building Nanosatellites, Formation & Evolution of Galaxies, Astronomical Software Engineering.

Università degli Studi di Roma "Tor Vergata"

Rome, Italy

MSC IN ASTROPHYSICS AND SPACE SCIENCE (ERASMUS MUNDUS JOINT MASTER'S PROGRAMME)

Sept. 2024 – Mar. 2025

- Completed first semester of the Erasmus Mundus Joint Master's in Astrophysics and Space Science.
- Key modules:** Quantum Mechanics for Astrophysics, Mathematical Methods, Radiative Processes, Modern Astrophysics, Numerical Methods.

University College London (UCL)

London, United Kingdom

MSci IN PHYSICS (FIRST CLASS HONOURS)

Sept. 2018 – Jun. 2022

- Thesis:** *Hunting for Intermediate-Mass Black Hole Binaries – or the Fun of Looking for the Needle in a Haystack.*
Supervisor: Prof. Daisuke Kawata (Mullard Space Science Laboratory).
- Graduated with a final average of 73% (First Class Honours) and a strong focus on Astronomy and Astrophysics. Conducted practical observing sessions and assisted in public outreach at the UCL Observatory in Mill Hill, London.
- Key modules:** General Relativity, Statistical Physics, Nuclear and Particle Physics, Solid State Physics, Quantum Mechanics, Mathematical Methods, Computing for Physics, Advanced Cosmology, Exoplanets, Galaxy Dynamics, Formation & Evolution, High-Energy Astrophysics.

"Ovidius" Theoretical High School

Constanța, Romania

ROMANIAN BACCALAUREATE, "PROFIL REAL" – MATHEMATICS-COMPUTER SCIENCE (BILINGUAL ENGLISH TRACK)

Sept. 2014 – Jun. 2018

- Graduated with a perfect 10/10 GPA; awarded *Diplomă de Bacalaureat* with final grade 9.87/10.

Research Experience

Observatoire de la Côte d'Azur (OCA)

Nice, France

MASTER'S THESIS: CHARACTERISING M-DWARF STARS AS EXOPLANET HOSTS – PREPARATION FOR THE PLATO MISSION

Sept. 2025 – present

- Supervised by Dr. Orlagh Creevey (OCA) and Dr. Ulrike Heiter (Uppsala University). Conducted as part of the Erasmus Mundus Joint Master's in Astrophysics and Space Science (MASS).
- Project within the ESA PLATO mission consortium, aimed at improving the precision of stellar parameters for low-mass exoplanet host stars.
- Deriving luminosities and radii of benchmark M-dwarf stars through spectral energy distribution (SED) fitting and bolometric flux calculations using *Gaia* and multi-band photometric data.
- Developing and testing Python-based analysis pipelines for photometric and spectroscopic data comparison, integrating literature data for validation.
- Contributing to the preparation and follow-up work of the PLATO mission by refining host-star models used for exoplanet characterisation.

Astrophysics Research Institute, Liverpool John Moores University

Liverpool, United Kingdom

SUMMER RESEARCH PROJECT: WHITE DWARF COOLING SEQUENCES AND INFRARED COLOUR EXCESS

Jul. 2025 – Sept. 2025

- Investigated whether the apparent broadening of the white dwarf cooling sequence in NGC 6397 is caused by an intrinsic IR-excess population or statistical scatter.
- Supervised by Prof. Maurizio Salaris; analysed JWST and HST photometric data of NGC 6397 to study the white dwarf cooling sequence.
- Constructed synthetic white dwarf populations using theoretical isochrones and a Salpeter IMF, introducing uncertainties via MC simulations.
- Developed Python routines to compare observed and simulated colour-magnitude diagrams, applying Gaussian Mixture Models (GMM) and Bayesian Information Criterion (BIC) analysis to assess multiple populations.
- Concluded that the observed colour spread is consistent with statistical noise, with no significant evidence for a distinct IR-excess population.

University College London

London, United Kingdom

RESEARCH INTERNSHIP: GALACTIC KINEMATICS AND STELLAR POPULATIONS WITH GAIA DR3

Jul. 2022 – Sept. 2022

- Supervised by Dr. Jason Sanders. Conducted exploratory analysis of *Gaia* DR3 data to investigate correlations between stellar kinematics, chemistry, and age in the Galactic disc.
- Applied basic machine learning techniques for stellar age estimation and used SQL queries for database access and data selection.
- Gained experience in large-scale astronomical data handling, model-data comparison, and Python-based analysis.

University College London

London, United Kingdom

MSCI THESIS: HUNTING FOR INTERMEDIATE-MASS BLACK HOLE BINARIES – “THE FUN OF LOOKING FOR A NEEDLE IN A

Oct. 2021 – Mar. 2022

HAYSTACK”

- Supervised by Prof. Daisuke Kawata (Mullard Space Science Laboratory). Investigated the detectability of intermediate-mass black holes (IMBHs) through their astrometric influence on stellar companions.
- Developed theoretical models for non-interacting black hole–star binaries and implemented them using the **Astromet** Python package.
- Computed orbital constraints from Keplerian dynamics and tidal radii, generating astrometric tracks for various IMBH–star configurations.
- Performed model fitting and residual analysis to evaluate *Gaia*’s capability to detect such systems, with implications for future IR astrometric missions.

University College London

London, United Kingdom

GROUP PROJECT: VOC ANALYSIS FOR DISEASE DETECTION – A CRITICAL ASSESSMENT OF NON-INVASIVE SCREENING METHODS

Jan. 2021 – Mar. 2021

- Collaborative third-year physics project (final grade: 76.8%) investigating the diagnostic potential of volatile organic compounds (VOCs) for non-invasive disease detection.
- Supervised by Prof. Gaetana Laricchia; reviewed detection methods including ion mobility spectrometry, electronic noses, and MALDI.
- Analysed biomarkers for diseases such as cancer, Alzheimer’s, and COVID-19; proposed an improved device concept for rapid VOC detection.
- Produced a group report, scientific poster, and final presentation summarising key findings and proposals.

Professional Experience

Northeastern University London

London, United Kingdom

RESIDENTIAL LIFE COORDINATOR

Sept. 2023 – Sept. 2024

- Full-time professional role supporting the wellbeing and academic success of first-year and study-abroad students.
- Served as the primary contact for 30–50 residents, providing pastoral, academic, and emotional support.
- Planned and implemented community engagement programmes, including orientation and cultural events, as per the Residential Curriculum.
- Handled emergency response and wellbeing interventions as part of the 24/7 on-call rotation, escalating cases to Residence Life Management.
- Collaborated with Student Affairs on arrival, move-out, and university-wide events to foster a safe, inclusive living environment.

International Hall, University of London

London, United Kingdom

RESIDENT ADVISOR

Aug. 2021 – Sept. 2022

- Supported the residential community of over 400 students by providing on-call assistance, welfare guidance, and peer mentorship.
- Organised social and cultural events to promote engagement and inclusivity among residents.
- Mediated conflicts, enforced residence policies, and coordinated with the Warden and administrative team on welfare issues.
- Assisted with move-ins, inductions, and end-of-year logistics to ensure smooth hall operations.

Academic Outreach & Teaching Experience

Department of Physics & Astronomy, University College London

London, United Kingdom

OUTREACH INTERN

Jun. 2022 – Sept. 2022

- Organised and coordinated outreach activities within local schools and community groups to promote physics and astronomy engagement.
- Planned and delivered science communication events highlighting departmental research, including participation in UCL’s public science festival *Your Universe*.
- Developed communication materials aimed at translating complex astrophysical topics for non-specialist audiences.

Eastbury Community School (Ogden Trust “Teach Physics” Programme)

Barking, United Kingdom

SUMMER PHYSICS TEACHING INTERN

Jun. 2022 – Jul. 2022

- Taught A-level and GCSE physics content through interactive sessions and lab-based exercises.
- Created and adapted classroom materials aligned with UK physics curricula.
- Assisted teachers in running lessons, monitoring student progress, and supporting extracurricular Science Club activities.

Access & Widening Participation Office, University College London

London, United Kingdom

STUDENT LEADER

Aug. 2022

- Supervised and supported students during a week-long residential Physics Summer School for Year 12 pupils from underrepresented backgrounds.
- Ensured student welfare and safety throughout the programme, acting as the primary point of contact for pastoral and academic support.
- Facilitated educational and social activities designed to encourage university participation in STEM disciplines.

UCL Observatory

Mill Hill, United Kingdom

DEMONSTRATOR

Oct. 2019 – Sept. 2022

- Trained to operate robotic and manual telescopes, including 14-inch Schmidt–Cassegrain and twin 24/18-inch refractors.
- Led public tours and observing sessions for amateur astronomers, school groups, and members of the public.
- Explained astronomical phenomena and instrumentation, developing strong science communication and public engagement skills.

University Employment & Mentoring Experience

Information Services Division, University College London (UCL)

London, United Kingdom

ROOM MONITOR

Oct. 2021 – Feb. 2022

- Provided on-site technical support to UCL teaching staff during lectures and seminars.
- Assisted with classroom setup, audiovisual equipment, and live teaching support across multiple departments.
- Ensured smooth operation of hybrid teaching environments and student engagement during sessions.

Department of Physics & Astronomy, University College London (UCL)

London, United Kingdom

SENIOR TRANSITION MENTOR

Sept. 2020 – Sept. 2022

- Supervised and supported a team of nine peer mentors, coordinating student transition and wellbeing programmes.
- Provided academic and pastoral guidance to first-year students, facilitating their integration into university life.
- Previously served as Transition Mentor (2019–2020), supporting 14 students through weekly meetings and individual advising.

Department of Physics & Astronomy, University College London (UCL)

London, United Kingdom

STUDENT AMBASSADOR

Nov. 2018 – Sept. 2022

- Represented the department during outreach events such as Open Days, campus tours, and school visits for prospective students.
- Engaged with A-level students and visitors, providing information about studying physics and life at UCL.

Leadership, Volunteering & Outreach

Oxford for Romania Foundation

Oxford, United Kingdom

VOLUNTEER – OXFORD FOR ROMANIA SUMMER SCHOOL

Aug. 2022 – Sept. 2022

- Supported the delivery of the *Oxford for Romania Summer School*, an educational programme empowering Romanian high-school students with limited access to higher education opportunities.
- Assisted with logistics, student management, and academic sessions, helping to foster a dynamic and inclusive learning environment.

Asociația 11even

Cluj-Napoca, Romania

FOUNDER – TEACHU.RO / CREATORI DE VIITOR

2017 – 2022

- Co-founded **TeachU.ro**, the first free online tutoring platform in Romania, launched under the *Champions United* educational initiative.
- Organised and coordinated volunteer tutors to offer free lessons to high-school students from underprivileged backgrounds.
- Contributed to platform development, online course materials, and outreach activities promoting equal access to education.
- Featured by national media as an example of youth-led educational innovation in Romania.

Astronomy and Astrophysics Club

Constanța, Romania

CO-FOUNDER AND TUTOR

Sept. 2016 – Sept. 2022

- Founded the Astronomy and Astrophysics Club to promote interest in astronomy among high-school students in Constanța.
- Organised annual contests and weekly meetings, and prepared students for the Romanian National Astronomy Olympiad.
- Developed and taught introductory and advanced astronomy courses for students of varying levels.

Engineers Without Borders, University College London

London, United Kingdom

TREASURER AND WORKSHOP EXECUTIVE

Jun. 2020 – Sept. 2021

- Managed the society's annual budget, liaised with the Students' Union on funding, and monitored financial operations throughout the year.
- Previously served as Workshop Executive (2019–2020), designing and running practical engineering workshops to encourage hands-on learning and teamwork.

Asociația 11even

Cluj-Napoca, Romania

VOLUNTEER – CHAMPIONS UNITED / CREATORI DE VIITOR

2019 – 2021

- Contributed to the organisation of *Creatori de viitor – Champions United*, a national educational programme fostering leadership and innovation among high-school students.
- Managed student groups, logistics, and event coordination during multiple editions, supporting workshops and mentor-led activities promoting teamwork and personal development.

National Astronomy and Astrophysics Student Camp, “Ștefan cel Mare” University

Suceava, Romania

MENTOR

Jan. 2020

- Selected to mentor participants at Romania’s National Astronomy and Astrophysics Camp, despite not holding formal teaching qualifications.
- Supervised and guided over 20 high-school students through advanced astronomy workshops, problem-solving sessions, and presentations.

Programme Committees

2024–2025 **Student Representative**, Quality Evaluation Committee - MASS

Rome

Scholarships & Honours

Erasmus Mundus Joint Master’s (EMJM) Scholarship, Awarded by the European Education and Culture

2024–2026 Executive Agency (EACEA) of the European Union for the Erasmus Mundus Master’s in Astrophysics and Space Science (MASS).

€33,600

2022 **UCL Summer Research Internship Scholarship**, Awarded by the Department of Physics & Astronomy, University College London, to support a research project on Milky Way formation and evolution.

£2,000

2022 **Ogden Trust “Teach Physics” Internship**, Selected for the national Ogden Trust programme supporting aspiring physics educators; completed teaching placement at Eastbury Community School, London.

£1,400

2018–2022 **UCL EU Bursary**, Awarded for academic merit and financial need to support four-year MSci Physics studies at University College London. Annual grant of approximately £2,500 per year.

£10,000

Skills & Professional Memberships

Programming Languages

Python, MATLAB, IDL

Scientific Tools & Software

TOPCAT, NASA SAO DS9, AstrolmageJ, Jupyter, Google Colab

Operating Systems & Environments

UNIX/Linux (Ubuntu), macOS, Windows, Git version control, LaTeX, MS Office

Languages

English: Fluent | **Romanian:** Native | **French:** Intermediate | **Italian:** Beginner

Professional Memberships & Societies

Mensa International; Institute of Physics (IOP);

England Handball Referee Association

Sports & Extracurricular Activities

England Handball Association

London, United Kingdom

HANDBALL REFEREE

Dec. 2023 – present

- Officiate national and regional handball and beach handball matches under the England Handball Association.
- Maintain high standards of fairness, communication, and professionalism on court, ensuring player safety and respect during competition.

UCL Handball Club

London, United Kingdom

COACH AND FORMER CAPTAIN

Oct. 2018 – Jun. 2024

- Served as President (2020–2021), Captain (2019–2020), and Vice-Captain (2018–2019), managing club activities for over 50 members.
- Coached and supported training sessions and matches, leading the team to success in national university competitions, including a bronze medal in 2022.

Islington Handball Club

London, United Kingdom

PLAYER AND VOLUNTEER

Oct. 2023 – Sep. 2024

- Compete in league and cup matches representing the club across multiple divisions.
- Assist with refereeing, coaching, and team logistics, contributing to the club’s operations and community engagement.